UNIVERSAL DATA CUSTOMISATION & AUTOMATION STUDIO

Complete Technical User Manual, System Architecture Blueprint & Dataset Use Case Compendium

Specification Attribute	System Configuration
Platform Release Version	v9.0.0 (Production Stable & Cloud Ready)
Core Architecture	Modular Python 3.13 / Streamlit Enterprise Multi-Page Router
Engineering Lead	Antigravity Engineering Platform Architecture Team
Supported Ingestion Formats	Excel (.xlsx, .xls), CSV, TSV, XML, JSON, TXT, ZIP Archives
Supported Export Formats	Excel (.xlsx), CSV, XML, JSON, TXT, Executive ZIP Package
Included Decoupled Modules	Universal XML Studio & TallyPrime Bulk Voucher Reconciliation Engine
Built-in AI & NLP Engine	Universal Semantic Heuristic NLP (100% Offline) + Optional Google Gemini LLM
Verification Benchmark	92 Unit & Integration Tests Passing (100% Automated Test Coverage)

MANUAL EXECUTIVE SUMMARY:

The Universal Data Customisation & Automation Studio is a general-purpose, enterprise-grade data engineering and automation platform. Unlike single-purpose converters or closed accounting add-ons, this studio adapts dynamically to any tabular, hierarchical, or delimited data source. It provides deep data health profiling, interactive column and row transformations, formula evaluations, selective record matching, in-place XML hierarchy editing, double-entry TallyPrime voucher reconciliation, sequential workflow pipeline authoring, AI-assisted natural language customisation, and multi-format export without ever mutating baseline original files.

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Chapter 1: System Architecture & Core Principles

The studio is architected around 4 non-negotiable enterprise data safety principles:

1. **Strict Baseline Immutability:** Original uploaded files are placed in an isolated, read-only memory register. All transformations, calculations, and deduplications occur exclusively on working copies.
2. **Bi-Directional Session Undo / Redo:** Every action produces an immutable snapshot in the change history stack. Users can traverse backward and forward across all steps without losing data.
3. **Zero Column Hallucination:** The AI Copilot and NLP engine use fuzzy schema matching against real column headers. The system rejects or clarifies ambiguous instructions rather than inventing fields.
4. **Decoupled Accounting Domain Engine:** TallyPrime voucher structures are handled in a dedicated module, ensuring accounting double-entry rules (`Debit == Credit`) do not pollute universal data tasks.

Chapter 2: Universal Ingestion & Automated Profiling

Located across pages/upload.py and pages/inspector.py, the universal ingestion engine automatically identifies file formats, detects encoding, resolves multi-sheet Excel files, and profiles data health.

Automated Data Profiling Telemetry:

- **Schema & Type Invariants:** Scans each column to determine true logical data types (numeric, datetime, boolean, string) and detects mixed data types within the same column.
- **Missingness Percentage:** Computes null, NaN, and whitespace counts per column, highlighting columns exceeding the 30% missingness threshold.
- **Duplicate Detection:** Identifies exact duplicate rows across all columns or composite key subsets with safety checks for unhashable objects (e.g. nested lists in JSON).
- **Date Pattern Anomaly Scanner:** Evaluates date strings against 8 common enterprise formats, flagging columns with mixed patterns (e.g. YYYY-MM-DD mixed with DD/MM/YYYY).
- **Statistical Outlier Detection:** Computes Interquartile Range (IQR) bounds [Q1 - 1.5*IQR, Q3 + 1.5*IQR] for numeric metrics.

Data Health Scoring Formula (0 to 100 Index):

```
Health Score = 100 - min(40, Missing% * 1.5) - min(30, Duplicate% * 1.5) - (DateAnomalies * 10) - (Outliers * 5)
```

Letter grades are mapped as: **A+** (95-100), **A** (85-94), **B** (75-84), **C** (65-74), **D** (55-64), and **F** (<55).

Supported File Format Matrix:

File Format	Extension	Max Upload	Internal Parser	Special Capabilities
Excel Workbook	.xlsx, .xls	200 MB	openpyxl / xlrd	Multi-sheet discovery, dynamic header detection, active sheet selector.
Comma Separated	.csv, .tsv	200 MB	pandas C-engine	Automatic delimiter detection (comma, tab, semicolon, pipe).
XML Document	.xml	200 MB	ElementTree / lxml	Lossless tree parsing, tag hierarchies, attribute retention, visual tree.
JSON Document	.json	200 MB	json / pandas	Record arrays, nested objects, tabular normalization.
Delimited Text	.txt	200 MB	built-in / pandas	Fixed-width tables, custom delimiters, log streams.
ZIP Archive	.zip	200 MB	zipfile	Automated unzipping and cataloging of contained supported files.

USE CASE 1: Profiling Dirty Sales & Logistics Dataset

Scenario: A business user receives a regional sales Excel workbook (sample_sales.xlsx) containing mixed dates, accounting formatted currencies, trailing whitespace, and duplicate records.

Table 1.1: Raw Input Dataset Preview (First 5 Rows):

OrderID	Customer	Region	Amount	OrderDate
1001	Acme Corp	North	$1,200.50	2026-01-15
1002	Globex Inc	South	2500.00	16/01/2026
1003	Initech	East	€150.75	20260117
1004	Hooli	West	(500.00)	2026-01-18
1004	Hooli	West	(500.00)	2026-01-18

Table 1.2: Automated Inspection Telemetry Results:

Audit Category	Detected Finding / Metric	Severity	Impact & Recommendation
Duplicate Records	OrderID '1004' appears twice (exact duplicate row).	HIGH	20% duplicate rate. Dedup required before posting.
Mixed Date Formats	'OrderDate' has 3 distinct formats (ISO, EU, Tally).	HIGH	Conflicting patterns: YYYY-MM-DD, DD/MM/YYYY, YYYYMMDD.
Dirty Numerics	'Amount' contains currency symbols ($ €) and parentheses.	MEDIUM	Stored as string (object). Cannot perform math operations.
Text Whitespace	'Customer' has leading/trailing padding (' Acme Corp ').	LOW	Text standardisation required for accurate grouping.

Table 1.3: Data Health Scorecard Output:

Health Metric	Calculated Score	Grade	Status Evaluation
Data Health Index	65.0 / 100	Grade C	Actionable anomalies detected. Cleaning recommended.

Chapter 3: Data Customisation & Transformation Studio

Located at pages/customisation.py, this module provides 6 specialized tabs for all data manipulation tasks:

1. Tab 1: ■ AI Natural Language Copilot

Translates plain English instructions into validated operations. Features 1-click prompt chips, live sandbox preview table, destructive operation warnings, and human confirmation gates.

2. Tab 2: ■ Column Operations

- **Rename Columns:** Rename single or multiple columns with collision prevention.
- **Add Column:** Constant values, auto-increment sequences, or column copies.
- **Delete & Reorder Columns:** Drop unnecessary fields and drag-reorder column order.
- **Merge & Split Columns:** Merge columns with custom delimiters or split strings into multiple fields.
- **Cast Column Types:** Safe type conversion to string, integer, float, datetime, or boolean.
- **Calculate Column (Formulas):** Mathematical expressions referencing columns in brackets, e.g. [Price] * [Qty] * 1.18 or [Revenue] - [Cost].
- **Map Column Values:** Value dictionary lookups (e.g. mapping state codes `NY` $\rightarrow$ `New York`).

3. Tab 3: ■ Row Operations

- **Filter Rows:** Dynamic conditional filtering (equals, contains, starts_with, ends_with, >, <, is_null, is_not_null, in_list).
- **Remove Duplicates:** Deduplicate across all columns or selected key subsets (keep first, last, or none).
- **Multi-Column Sorting:** Hierarchical sorting across multiple columns with ascending/descending controls.
- **Row Slicing:** Head, tail, or random sample record selection.

4. Tab 4: ■ Data Cleaning & Quality

- **Impute Missing Values:** Fill nulls via constant, mean, median, mode, forward-fill, back-fill, or drop incomplete rows.
- **Standardize Text:** Strip leading/trailing whitespace, collapse multi-spaces, convert to UPPERCASE, lowercase, or Title Case, strip punctuation.
- **Standardize Dates:** Parses mixed date representations (YYYY-MM-DD, DD/MM/YYYY, MM/DD/YYYY, YYYYMMDD) into target format.
- **Clean Numeric Formatting:** Strips currency symbols ($ € £ ■ ¥), commas, spaces, percentage signs, and converts accounting parentheses (500) into -500.0.
- **Treat Outliers:** Detects IQR/Z-score outliers and caps them to boundaries, flags them, or replaces them with nulls.

5. Tab 5: ■ Reshape & Multi-Dataset

- **Dataset Joins:** Inner, left, right, and outer joins matching on key columns.
- **Append Datasets:** Stack multiple tabular datasets vertically with automatic schema alignment.
- **Pivot & Unpivot:** Pivot long data into summary tables, or melt wide tables into tall attribute-value rows.

USE CASE 2: Sales Register Sanitation & 18% GST Calculation

Scenario: Clean the dirty sales dataset from Use Case 1, remove duplicates, normalize dates, calculate 18% GST, and aggregate total revenue by region.

Step-by-Step Studio Execution Plan:

1. **Clean Currency Strings:** Navigate to ■ *Data Cleaning* $\rightarrow$ *Clean Numeric Formatting*. Select column Amount, check *Strip Currency & Symbols*. Converts $1,200.50 $\rightarrow$ 1200.50 and (500.00) $\rightarrow$ -500.00.
2. **Standardize Dates:** Under *Standardize Date Formats*, select OrderDate with target format %Y-%m-%d. Resolves 16/01/2026 and 20260117 to standard ISO format.
3. **Trim Text:** Under *Standardize Text*, select Customer with operations [trim, titlecase].
4. **Remove Duplicate Orders:** Under ■ *Row Operations* $\rightarrow$ *Remove Duplicates*, select key column OrderID, keep mode: first.
5. **Calculate Tax Formula:** Under ■ *Column Operations* $\rightarrow$ *Calculate Column / Formula*, enter Target: TotalWithTax and Formula: [Amount] * 1.18.

Table 2.1: Transformed Clean Output Dataset:

OrderID	Customer	Region	Amount	OrderDate	TotalWithTax
1001	Acme Corp	North	1200.50	2026-01-15	1416.59
1002	Globex Inc	South	2500.00	2026-01-16	2950.00
1003	Initech	East	150.75	2026-01-17	177.88
1004	Hooli	West	-500.00	2026-01-18	-590.00

Table 2.2: Regional Group-By & Aggregation Summary (Step 6):

Region	Total Amount (Base)	Total With GST (18%)	Order Count
East	150.75	177.88	1
North	1200.50	1416.59	1
South	2500.00	2950.00	1
West	-500.00	-590.00	1

Health Score Improvement: Following these operations, the Data Health Score increases from **65.0 (Grade C)** to **100.0 (Grade A+)** with zero duplicate rows and zero date anomalies.

Chapter 4: Advanced Record Matching & Reconciliation

Located at pages/matching.py, this engine compares an **Original Dataset** against a **Revised Dataset** across composite keys, isolates cell deltas, and executes selective reconciliation.

The 4-Category Record Classification Model:

When comparing datasets, every record is placed into an unambiguous mutual category:

Category	Mathematical Definition	Visual Status	System Reconciliation Policy
IDENTICAL	Key in both; all compared attributes equal.	■ IDENTICAL	Preserved as pristine. No modification needed.
MODIFIED	Key in both; one or more cell values differ.	■ MODIFIED	Side-by-side cell delta display. Update selected fields.
NEW	Key exists only in revised dataset.	■ NEW	Staged for insertion into master dataset.
MISSING	Key exists only in original; omitted in revised.	■ MISSING	Flagged as deleted/omitted. Protected from accidental deletion.

Selective Reconciliation Execution Policies:

- **Update Existing Modified Records:** Overwrites designated attributes in common records with revised values.
- **Insert New Records:** Appends fresh records found in revised dataset into the reconciled output.
- **Remove Missing Records:** *(Off by default)* Requires explicit confirmation to prevent accidental loss of omitted records.

USE CASE 3: Purchase Order Master Reconciliation

Scenario: Compare Master Order Register against Revised Vendor Invoice Register based on OrderID.

Table 3.1: Original Master Register vs Revised Vendor Register:

OrderID	Original Customer	Orig Amount	Orig Status	Revised Customer	Rev Amount	Rev Status
1001	Acme Corp	1500.00	Pending	Acme Corp	1750.00	Completed
1002	Globex	840.00	Pending	Globex Inc	840.00	Pending
1003	Initech	2300.00	Delivered	Initech	2300.00	Delivered
1004	Hooli	950.00	Pending	— (Missing)	—	—
1006	— (New)	—	—	Pied Piper	4800.00	Delivered

Table 3.2: Side-by-Side Cell Delta Breakdown (Engine Output):

OrderID	Match Status	Changed Fields	Original Value	Revised Value
1001	MODIFIED	Amount, Status	Amount: 1500.00 | Status: Pending	Amount: 1750.00 | Status: Completed
1002	MODIFIED	Customer	Customer: Globex	Customer: Globex Inc
1003	IDENTICAL	None	All fields identical	All fields identical
1004	MISSING	All Fields	Active in Master Register	Omitted from Vendor Statement
1006	NEW	All Fields	Not in Master Register	New Vendor Invoice Addition

Table 3.3: Final Reconciled Output Dataset (Policy: Update + Insert + Retain Missing):

OrderID	Customer	Amount	Status	Audit Reconciliation Note
1001	Acme Corp	1750.00	Completed	Updated: Amount adjusted by +250.00; Status changed.
1002	Globex Inc	840.00	Pending	Updated: Customer legal name updated.
1003	Initech	2300.00	Delivered	Unmodified: Preserved from master baseline.
1004	Hooli	950.00	Pending	Retained: Preserved missing record from deletion.
1006	Pied Piper	4800.00	Delivered	Inserted: Newly added vendor record.

Chapter 5: Universal XML Studio & Tabular Mapping

Located at pages/xml_studio.py, this module provides an enterprise-grade XML manipulation IDE and tabular-to-XML mapping engine.

Key XML Studio Capabilities:

- **Hierarchical Node Search:** Search deeply nested XML structures by tag name, attribute key/value, or inner text.
- **In-Place Lossless Mutation:** Edit tag text, update attribute values, add child elements, or remove specific elements while preserving namespaces, encoding, and comments.
- **Tabular-to-XML Mapping Engine:** Map flat Excel/CSV columns into complex nested XML structures with attributes and sub-elements.
- **Cross-Format Conversions:** 1-click lossless conversions between Excel, CSV, XML, and JSON with syntax validation before export.

USE CASE 4: Mapping Flat Customer CSV to Hierarchical XML

Scenario: Convert a flat Customer table into a hierarchical XML document with nested geo and financial nodes.

Table 4.1: Source Customer Tabular Dataset:

CustID	CustomerName	City	Balance
CUST-001	Acme Corporation	Boston	45000.00
CUST-002	Globex Industries	New York	12500.50

Tabular-to-XML Mapping Rules Configured in UI:

```
Root Tag: CUSTOMERS | Record Tag: CUSTOMER • CustID $\rightarrow$ @id (XML Attribute) • CustomerName $\rightarrow$ DETAILS/NAME (Nested Tag) • City $\rightarrow$ LOCATION/CITY (Nested Tag) • Balance $\rightarrow$ FINANCIAL/OUTSTANDING_BALANCE (Nested Tag)
```

Generated Output XML Document:

```
<CUSTOMERS>
<CUSTOMER id="CUST-001">
<DETAILS><NAME>Acme Corporation</NAME></DETAILS>
<LOCATION><CITY>Boston</CITY></LOCATION>
<FINANCIAL><OUTSTANDING_BALANCE>45000.00</OUTSTANDING_BALANCE></FINANCIAL>
</CUSTOMER>
<CUSTOMER id="CUST-002">
<DETAILS><NAME>Globex Industries</NAME></DETAILS>
<LOCATION><CITY>New York</CITY></LOCATION>
<FINANCIAL><OUTSTANDING_BALANCE>12500.50</OUTSTANDING_BALANCE></FINANCIAL>
</CUSTOMER>
</CUSTOMERS>
```

Chapter 6: Dedicated TallyPrime Bulk Voucher Updater

Located at pages/tally_updater.py, this specialized module reconciles and updates vouchers in TallyPrime XML export files without destroying complex accounting allocations.

Why Specialized Tally Support is Mandatory:

TallyPrime XML documents contain deeply nested structures: GUIDs, RemoteIDs, GST details, inventory entries, bill allocations, cost centres, and bank ledger records. Naive flat-table conversions erase these nested nodes, rendering the XML un-importable. Our engine performs in-place mutation, updating ledger amounts while preserving 100% of GUIDs and untouched elements.

The 4-Step Tally Updating Wizard:

1. **Upload Baseline Tally XML & Revised Excel:** Ingests Tally export XML and an Excel workbook containing revised ledger values.
2. **Match Vouchers:** Matches existing vouchers using GUID or VoucherNumber + Date. Stops execution if duplicate keys exist in the Tally XML.
3. **Double-Entry Balance Verification:** Strict accounting audit checking that Total Debit == Total Credit (Σ = 0). Flags unbalanced vouchers.
4. **Generate Alter XML & 4-Sheet Audit Report:** Stamps ACTION='Alter' on modified vouchers and generates a comprehensive 4-sheet reconciliation Excel workbook.

USE CASE 5: Bulk Modifying Tally Sales Vouchers

Scenario: Update ledger amounts for voucher VCH-2026-001 (GUID: 7a3c8e92-...) based on audited bank figures.

Table 5.1: Revised Excel Ledger Updates:

GUID / Identifier	LedgerName	Original Amount	Revised Amount	Accounting Type
7a3c8e92-1b4d-4e5f-9a1c-000000000001	National Trading Co.	118000.00	141600.00	Debit (Party Ledger)
7a3c8e92-1b4d-4e5f-9a1c-000000000001	Sales Account	-100000.00	-120000.00	Credit (Sales)
7a3c8e92-1b4d-4e5f-9a1c-000000000001	Output CGST	-9000.00	-10800.00	Credit (Tax)
7a3c8e92-1b4d-4e5f-9a1c-000000000001	Output SGST	-9000.00	-10800.00	Credit (Tax)

Table 5.2: Double-Entry Balance Audit Results:

Voucher No	GUID	Total Debits	Total Credits	Net Difference	Audit Status
VCH-2026-001	7a3c8e92-...	141,600.00	141,600.00	0.00	■ BALANCED

Generated Tally Import XML Snippet (Showing Stamped Alter Action):

```
<TALLYMESSAGE xmlns:UDF="TallyUDF">
<VOUCHER VCHTYPE="Sales" ACTION="Alter">
<GUID>7a3c8e92-1b4d-4e5f-9a1c-000000000001</GUID>
<VOUCHERNUMBER>VCH-2026-001</VOUCHERNUMBER>
<DATE>20260115</DATE>
<ALLLEDGERENTRIES.LIST>
<LEDGERNAME>National Trading Co.</LEDGERNAME>
<ISDEEMEDPOSITIVE>Yes</ISDEEMEDPOSITIVE>
<AMOUNT>-141600.00</AMOUNT>
<!-- Sub-allocations, Bill details, and GST preserved 100% -->
</ALLLEDGERENTRIES.LIST>
</VOUCHER>
</TALLYMESSAGE>
```

Chapter 7: Custom Workflow Builder & AI Copilot

1. Custom Workflow Builder ('Build My Workflow')

Located at pages/workflow_builder.py, this engine chains operations sequentially into reusable recipes. The output DataFrame of Step *N* automatically becomes the input to Step *N+1*.

USE CASE 6: Assembling a 5-Step Automated Recipe

Table 6.1: Sequential Pipeline Execution Telemetry:

Step #	Operation	Category	Input Shape	Output Shape	Duration	Status
Step 1	standardize_text	Data Cleaning	5 rows × 5 cols	5 rows × 5 cols	2.1 ms	■ SUCCESS
Step 2	clean_numeric	Data Cleaning	5 rows × 5 cols	5 rows × 5 cols	3.4 ms	■ SUCCESS
Step 3	remove_duplicates	Row Operations	5 rows × 5 cols	4 rows × 5 cols	1.8 ms	■ SUCCESS
Step 4	calculate_column	Column Operations	4 rows × 5 cols	4 rows × 6 cols	2.2 ms	■ SUCCESS
Step 5	assert_unique_key	Validation	4 rows × 6 cols	4 rows × 6 cols	0.9 ms	■ SUCCESS

2. Natural Language Customisation & AI Copilot (Phase 8)

Located inside *4. Data Customisation* (Tab 1) and *8. Workflow Builder*. Translates plain English instructions into structured operations with strict anti-hallucination schema grounding.

USE CASE 7: Natural Language Instruction Compilation

Table 7.1: Natural Language Prompts $\rightarrow$ Compiled Operations Matrix:

User Plain English Prompt	Compiled Operation	Target Params	Destructive?
'Rename Patient ID to Registration Number'	rename_columns	{'mapping': {'Patient ID': 'Registration Number'}}	No (Safe)
'Remove duplicate records by Invoice Number'	remove_duplicates	{'subset_cols': ['Invoice Number'], 'keep': 'first'}	Yes (Drops rows)
'Convert date column into DD-MM-YYYY'	standardize_dates	{'column': 'VisitDate', 'target_format': '%d-%m-%Y'}	No (Safe)
'Calculate Tax as [Amount] * 0.18'	calculate_column	{'target_col': 'Tax', 'formula': '[Amount] * 0.18'}	No (Safe)
'Drop column Status and Age'	delete_columns	{'columns': ['Status', 'Age']}	Yes (Deletes cols)
'Cap outliers in Amount'	treat_outliers	{'column': 'Amount', 'method': 'iqr', 'action': 'cap'}	No (Safe)

Human Confirmation Gate for Destructive Operations:

When an operation deletes rows or drops columns (e.g. remove_duplicates or delete_columns), the AI Copilot displays a high-visibility warning banner and requires explicit human confirmation via the ■ **Confirm & Execute Transformation** button after inspecting the live sandbox preview.

Chapter 8: Validation, Export & Cloud Deployment

1. Validation & Quality Centre (pages/validation.py)

Generates an executive 5-sheet OpenXML (`.xlsx`) diagnostic workbook:

- **Sheet 1: Executive_Summary** — Dataset dimensions, health score, grade, and finding counts.
- **Sheet 2: Data_Quality_Findings** — Categorized issues (Missing, Duplicates, Dates, Outliers) with severity and recommendations.
- **Sheet 3: Rule_Validation_Results** — Mandatory column existence, key uniqueness, and non-negative constraint audits.
- **Sheet 4: Transformation_History** — Chronological log of all session modifications with timestamp and row deltas.
- **Sheet 5: Problematic_Records** — Extracted sample of invalid rows (nulls, duplicates) for direct review.

2. Multi-Format Export Centre (pages/export.py)

Exports cleaned datasets into Excel (.xlsx), CSV (.csv), XML (.xml), JSON (.json), TXT (.txt), or an all-in-one ZIP package containing all 5 formats plus the 5-sheet Quality Audit Workbook.

3. Public Cloud Deployment Guide (Streamlit Community Cloud)

The repository is production-ready for instant free cloud hosting on Streamlit Community Cloud:

Step A: Push Project to GitHub (via GitHub Desktop or CLI)

1. Open **GitHub Desktop** (pre-installed on your computer). 2. Click **File** $\rightarrow$ **Add Local Repository...** and select C:\Users\HP\Documents\tally_edit. 3. Click the blue **'Publish repository'** button (uncheck 'Keep this code private' for free hosting).

Step B: Deploy Live on Streamlit Cloud

1. Open your browser and navigate to **share.streamlit.io** (sign in with GitHub). 2. Click **'New app'**. 3. Select repository: <YOUR-USERNAME>/tally_edit | Branch: main | Main file: app.py. 4. Click **'Deploy!'**. Streamlit Cloud reads requirements.txt, configures the environment, and launches the app worldwide!

End of Official Technical Manual — Universal Data Customisation & Automation Studio v9.0.0